



$$\frac{x}{y} = \frac{5}{7} \checkmark$$

$$\begin{array}{r} 2x + 3y \\ -x + 9y \\ \hline \end{array}$$

$$\begin{array}{l} m=2 \\ n=3 \end{array}$$

$$\begin{array}{l} p=7 \\ q=9 \end{array}$$

Given $x : y$,
find $(\underline{mx} + \underline{ny}) : (\underline{px} + \underline{qy})$ where
 m, n, p, q are constants

Que: If $x : y = 2 : 3$, find the value of $3x + 2y : 2x + 5y$?

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$$\frac{x}{y} = \frac{2}{3}$$

$$x = \frac{2y}{3}$$

$$\frac{3x + 2y}{2x + 5y}$$

$$\frac{4}{19}$$

$$\frac{4 \times 3}{19} = \frac{12}{19} \checkmark$$

$$\frac{3 \left[\frac{2y}{3} \right] + 2y}{2 \left[\frac{2y}{3} \right] + 5y}$$

$$\frac{3 \times \frac{2}{3} + 2}{2 \times \frac{2}{3} + 5} = \frac{4}{\frac{4}{3} + 5} = \frac{4}{\frac{4}{3} + \frac{15}{3}}$$

$$\frac{3 \times \frac{2}{3} + 2}{2 \times \frac{2}{3} + 5}$$

Que: If $x : y = 2 : 3$, find the value of $3x + 2y : 2x + 5y$?

$$\frac{x}{y} = \frac{2}{3}$$

$$\frac{3x+2y}{2x+5y}$$

$$\frac{\frac{3xy}{y} + 2y}{\frac{2xy}{y} + 5y}$$

$$\frac{3\frac{x}{y} + 2}{2\frac{x}{y} + 5}$$

$$y \left[3\frac{x}{y} + 2 \right]$$

$$y \left[2\frac{x}{y} + 5 \right]$$

$$\frac{3\left(\frac{2}{3}\right) + 2}{2\left(\frac{2}{3}\right) + 5}$$

$$= \frac{4}{\frac{4}{3} + 5}$$

$$= \frac{4}{\frac{4+15}{3}}$$

$$= \frac{4}{\frac{19}{3}}$$

$$\frac{12}{19} \checkmark$$

Que: If $x : y = 2 : 3$, find the value of $3x + 2y : 2x + 5y$?

$$\frac{x}{y} = \frac{2}{3}$$

$$\text{let } x = \frac{2}{3} \quad y = 1$$

$$\frac{x}{y} = \frac{2}{3} = \frac{2}{3}$$

$$\frac{3x + 2y}{2x + 5y}$$

$$\frac{3\left(\frac{2}{3}\right) + 2(1)}{2\left(\frac{2}{3}\right) + 5(1)}$$

$$= \frac{4}{\frac{4}{3} + 5}$$

$$= \frac{4}{\frac{4+15}{3}}$$

$$= \frac{4 \times 3}{19}$$

$$= \frac{12}{19}$$

Que: If $x : y = 2 : 3$, find the value of $3x + 2y : 2x + 5y$?

$$\frac{x}{y} = \frac{2}{3}$$

$$x = \frac{2}{3}, y = 1$$

let $x = 2$
 $y = 3$

$$\frac{x}{y} = \frac{2}{3} \quad x = 2k, y = 3k$$

$$\frac{3x + 2y}{2x + 5y} \Rightarrow \frac{3(2) + 2(3)}{2(2) + 5(3)} = \frac{6 + 6}{4 + 15} = \frac{12}{19} \checkmark$$

$$\frac{3(2k) + 2(3k)}{2(2k) + 5(3k)}$$

$$\frac{k(3 \times 2 + 2 \times 3)}{k(2 \times 2 + 5 \times 3)}$$

Que: If $x : y = 7 : 9$, find the value of $3x - 7y : 9x - 11y$?

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$$\frac{x}{y} = \frac{7}{9}$$

$$x = 7$$
$$y = 9$$

$$\frac{7}{9}$$

$$\frac{3x - 7y}{9x - 11y} = \frac{3(7) - 7(9)}{9(7) - 11(9)}$$

$$\frac{7}{9} \checkmark$$

$$\begin{array}{r} 99 \\ - 63 \\ \hline 36 \end{array}$$

$$= \frac{21 - 63}{63 - 99} = \frac{-42}{-36} = \frac{7}{6}$$